

SOP-VLV-100: Valve Inspection and Maintenance Procedure

Document owner: Mechanical Integrity Department **Effective date:** 2025-01-15
Review cycle: 24 months

Section 1. Purpose and Scope

This procedure defines the requirements for periodic inspection, testing, and maintenance of manual and actuated valves in hydrocarbon and utility service across all process units. It applies to gate, globe, ball, butterfly, check, and pressure relief valves rated Class 150 and above. It does not apply to instrument air valves smaller than 1 inch nominal pipe size, which are covered under SOP-INST-220.

Compliance with this procedure is mandatory for all mechanical technicians, inspectors, and contractors performing valve work. Deviations require written authorization from the Mechanical Integrity Lead.

Section 2. Inspection Intervals

Clause 2.1 — Critical service valves (defined as valves isolating pressure vessels above 10 barg or handling H₂S concentrations above 100 ppm) shall be visually inspected every 6 months and function-tested every 12 months.

Clause 2.2 — Non-critical service valves shall be visually inspected every 12 months and function-tested every 24 months.

Clause 2.3 — Pressure relief valves (PRVs) shall be bench-tested and recertified every 36 months, or every 12 months if in dirty or fouling service as determined by the process engineer.

Clause 2.4 — Any valve that fails a function test shall be re-tested within 30 days of corrective maintenance, and the failure shall be logged in the Valve Integrity Register with root cause noted.

Section 3. Visual Inspection Procedure

Step 3.1 — Confirm the valve is correctly tagged and matches the P&ID reference before beginning work. A mismatch shall be reported to the area supervisor immediately and work shall not proceed.

Step 3.2 — Inspect the valve body and bonnet for external corrosion, pitting, or leakage at the packing gland and flange connections. Any visible weeping shall be documented with a photograph and a severity rating of Minor, Moderate, or Severe.

Step 3.3 — Check actuator linkages, air lines, and solenoids on actuated valves for physical damage, loose fittings, or signs of corrosion. Actuator travel stops shall be verified against the original commissioning setpoint.

Step 3.4 — Verify handwheel or lever operation is smooth across the full range of travel with no excessive force, binding, or backlash exceeding 5 degrees.

Step 3.5 — Record all findings on Form MI-014 (Valve Visual Inspection Checklist) and submit to the Mechanical Integrity Department within 48 hours of inspection.

Section 4. Function Testing

Clause 4.1 — Function testing verifies full stroke operation from fully open to fully closed and confirms travel time is within 10 percent of the original design specification.

Clause 4.2 — For emergency shutdown (ESD) valves, stroke time shall not exceed 3 seconds for valves 6 inches and below, or 8 seconds for valves above 6 inches, unless a specific process safety time study dictates otherwise.

Clause 4.3 — Any valve exceeding the allowable stroke time by more than 20 percent shall be immediately tagged out of service and escalated to the Process Safety Engineer for a risk assessment before returning to operation.

Clause 4.4 — Function test results shall be entered into the Computerized Maintenance Management System (CMMS) within 24 hours, referencing the valve tag, test date, technician ID, and pass/fail status.

Section 5. Torque and Bolting Requirements

Step 5.1 — Bonnet and flange bolting shall be torqued in a star pattern to the value specified in Table T-2 of the Piping Bolting Standard, using a calibrated torque wrench with a valid calibration certificate dated within the last 12 months.

Step 5.2 — Gasket material shall match the service specified on the line class datasheet. Substitution of gasket material requires written approval from the Materials Engineer.

Step 5.3 — After bolt-up, a hydrostatic or pneumatic leak test shall be performed per SOP-PT-150 before the valve is returned to service.

Section 6. Escalation and Reporting

Clause 6.1 — Any valve found leaking to atmosphere at a rate that presents an immediate safety hazard shall trigger an immediate shutdown request through the Control Room and shall not wait for the standard 48-hour reporting window described in Section 3.

Clause 6.2 — Repeat failures (two or more failures within a 12-month period) on the same valve tag require a Root Cause Failure Analysis (RCFA) to be initiated by the Reliability Engineer within 5 working days.

Clause 6.3 — All inspection and test records shall be retained for a minimum of 10 years and are subject to audit during regulatory inspections.